

C.2 The wave model

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **27 marks**

Take $c = 3.0 \times 10^8 \text{ m s}^{-1}$. Show your working.

Question 1 — Radio Aotearoa

(6 marks)

A community radio station broadcasts at a frequency of 101 MHz.

- (a) Distinguish between transverse and longitudinal waves, giving one example of each. [2]
- (b) Calculate the wavelength of the station's radio waves. [2]
- (c) An astronaut's radio works perfectly on a spacewalk, yet if she taps a spanner against her helmet, a crewmate floating two metres away hears nothing. Explain both observations. [2]

Question 2 — Whale karaoke

(8 marks)

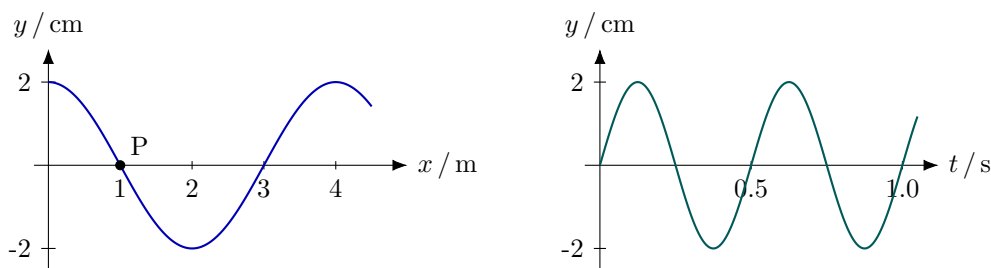
A humpback whale sings at a frequency of 25 Hz. Sound travels at 1500 m s^{-1} in sea water.

- (a) Calculate the wavelength of the song. [2]
- (b) Calculate the period of one oscillation. [1]
- (c) A second whale listens from 2 km away, then swims to 6 km away. Assuming the sound spreads uniformly in all directions, determine the factor by which the received intensity changes. [3]
- (d) A student watching a nature documentary says the whale's song "pushes the water all the way to the second whale". Explain what the wave actually transports, and what the water does. [2]

Question 3 — The seismometer trace

(9 marks)

The diagram shows two graphs recorded for the same transverse wave travelling to the *right* along a rope: a snapshot of displacement against position, and a displacement–time record for the single rope point P.



- (a) From the appropriate graph, write down the amplitude and the wavelength of the wave. [2]
- (b) From the appropriate graph, write down the period, and hence calculate the speed of the wave. [3]

- (c) Explain which physical quantities can *only* be read from the displacement–time graph, and why the snapshot cannot supply them. [2]
- (d) State the direction in which point P is moving at the instant of the snapshot, and justify it using the direction of wave travel. [2]

Question 4 — Half a period later

(4 marks)

At one instant, a particle in a transverse wave is at zero displacement and moving upwards.

- (a) State the particle’s displacement and direction of motion exactly half a period later. [2]
- (b) A classmate answers “at a crest”. Identify the time interval for which that answer *would* be correct, and hence the error made. [2]

Answers

Question 1 — Radio Aotearoa

- (a) *Transverse*: the oscillation is perpendicular to the direction of energy propagation. Examples: light, ripples on water, a wave on a rope. *Longitudinal*: the oscillation is parallel to the propagation direction. Examples: sound, and some seismic waves.
- (b) $\lambda = c/f = 3.0 \times 10^8 / 101 \times 10^6 = 3.0 \text{ m}$ (more precisely 2.97 m).
- (c) Radio is an electromagnetic wave and needs no medium, so it crosses the vacuum between them. Sound is mechanical and requires a medium; there is no air between the suits, so the tapping travels only through her own suit and helmet.

Question 2 — Whale karaoke

- (a) $\lambda = v/f = 1500/25 = 60 \text{ m}$.
- (b) $T = 1/f = 1/25 = 0.040 \text{ s}$.
- (c) Intensity obeys $I \propto 1/d^2$. The distance triples, from 2 km to 6 km, so the intensity falls by a factor $3^2 = 9$. It drops to one *ninth*, not one third.
- (d) The wave transports *energy* (and with it information). The water itself merely oscillates about a fixed position and ends up where it started.

Question 3 — The seismometer trace

- (a) Amplitude 2.0 cm and wavelength 4.0 m, both read from the *snapshot* (displacement against position).
- (b) Period 0.50 s, read from the displacement–time graph. Then $v = \lambda/T = 4.0/0.50 = 8.0 \text{ m s}^{-1}$.
- (c) The period, and hence the frequency. The snapshot freezes a single instant, so it cannot show how fast anything repeats; it carries distance information only. Only the time record can supply a period.
- (d) P is moving **upwards**. The wave travels to the right, so nudge the whole waveform slightly rightwards: P then takes the displacement currently held by the point just to its *left*, which on this descending slope is higher than P. (The algebra agrees: $\partial y/\partial t = -v \partial y/\partial x$, and the slope at P is negative, so $\partial y/\partial t > 0$.)

Question 4 — Half a period later

- (a) Zero displacement again, but moving *downwards*. After half a period the particle has returned to equilibrium with its motion reversed.
- (b) “At a crest” would be correct after a *quarter* of a period. The classmate has used quarter-period reasoning for a half-period question, confusing the time to reach maximum displacement with the time to complete half a cycle.